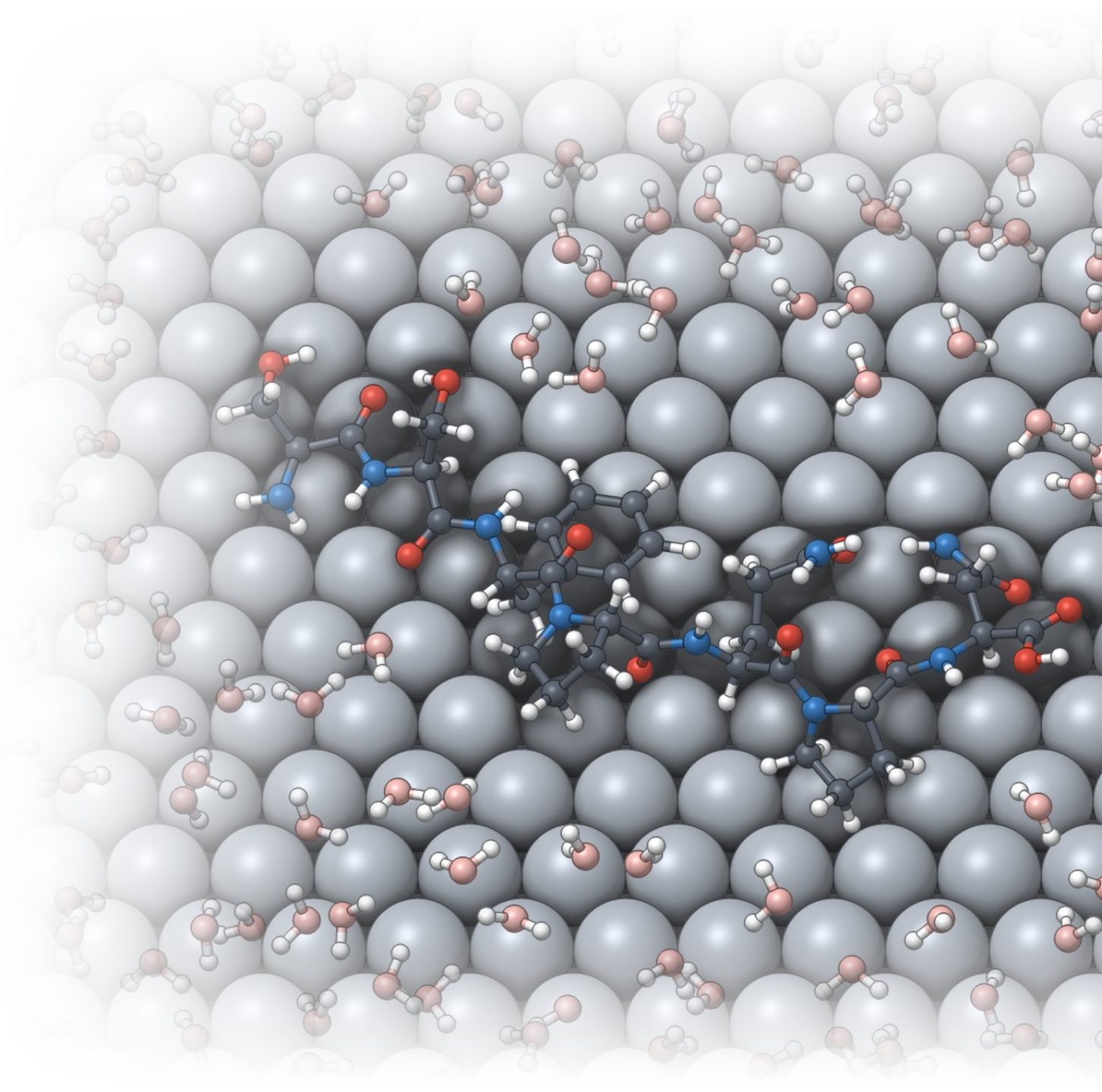
Interface Force Field

A beginner’s guide to simulating where molecules meet solid materials

The picture shows a peptide, a short chain of amino acids, lying on a platinum surface in water. This guide shows how a computer can work out where molecules like this stick, starting from a single pair of atoms.

Sean Florez

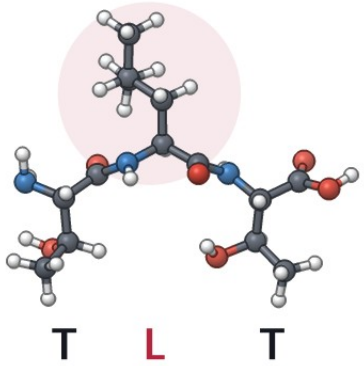


Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

Can one amino acid change the shape of a metal crystal?

A peptide is a short chain of amino acids, the building blocks of proteins. Ruan and colleagues grew platinum crystals in water with one peptide present. Changing only its middle amino acid from L to F, the one with a flat ring, shifted the crystals from cubes toward tetrahedra, pyramids with four triangular faces.

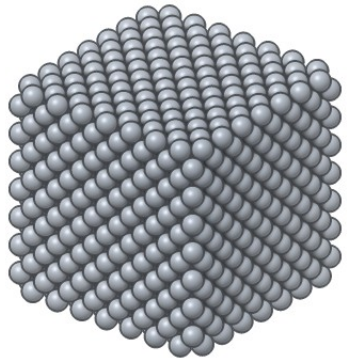
In these syntheses, molecules that stick to one kind of face slow its growth, so those faces grow slowest and end up covering the crystal. So why does F stick to the triangular faces?

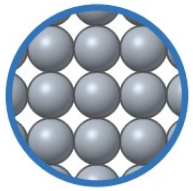


T L T

threonine · leucine · threonine

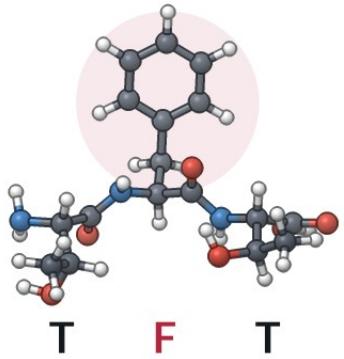
grows





Cubes

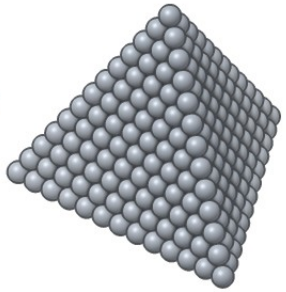
square pattern of atoms on every face

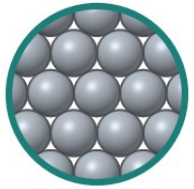


T F T

threonine · phenylalanine · threonine

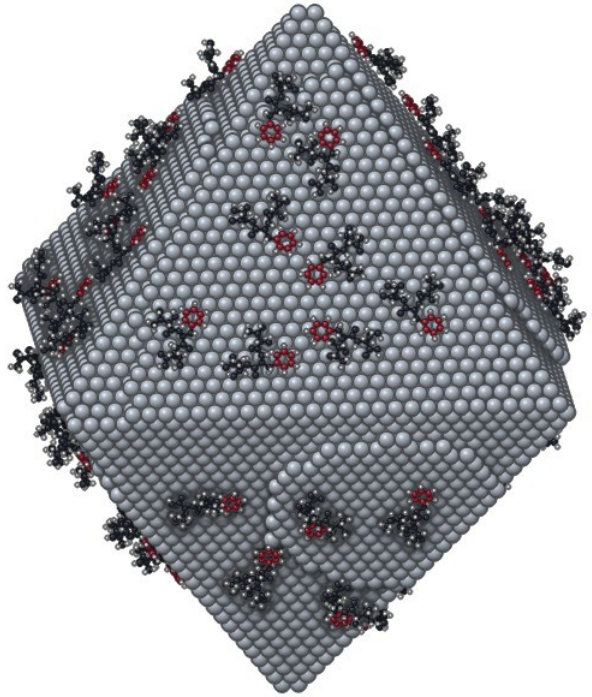
shifts toward





Tetrahedra

triangular pattern of atoms on every face



An illustration of the idea, not a simulation from the paper. Peptides slow growth on the triangular faces, so the square faces (blue) shrink and vanish, leaving an octahedron, which also has only triangular faces.

Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

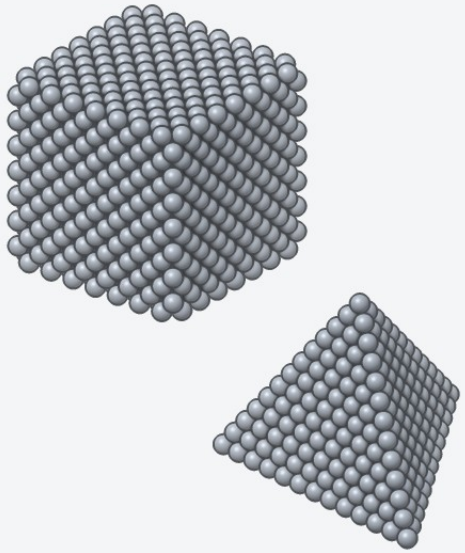
Where molecules stick changes how materials grow and work

The platinum puzzle is one case of a much wider question. Wherever a solid meets water and molecules, what sticks where changes how the material grows and behaves, from tiny metal crystals to bones and cement.

If a computer can get these contacts right, ideas can be tested before anything is made, and that is what IFF is built to do.

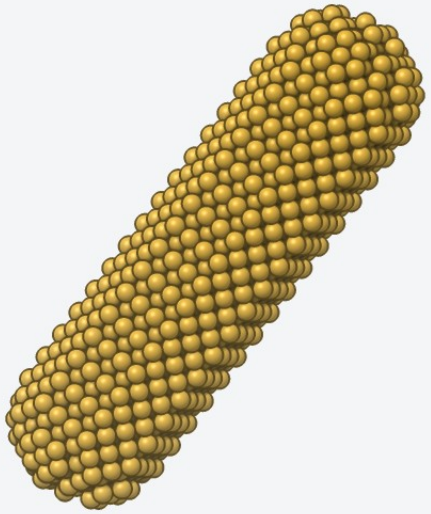
Catalysts

Platinum crystals speed up reactions. What sits on their faces affects their shape and how well they work.



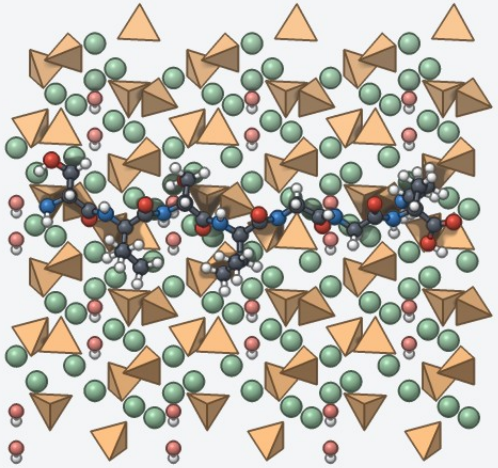
Medicine

Gold rods that turn light into heat are studied for treating cancer. Their surfaces interact with water and biomolecules.



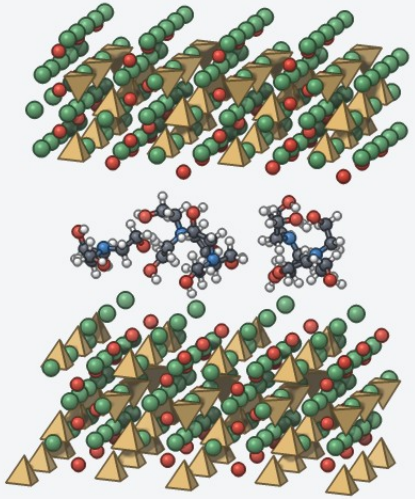
Bones and teeth

Proteins steer how the mineral of bones and teeth grows. That matters for osteoporosis and tooth decay.



Cement

Grinding is a big cost in making cement. Additives that stick to the grains stop them clumping, so it takes less time and energy.



Ruan et al., Nano Lett. (2013) • Geada et al., Nat. Commun. (2018) • Lin and Heinz, J. Phys. Chem. C (2016) • Mishra et al., J. Phys. Chem. C (2013) • Patent EP2527307A1

Models built separately do not automatically fit together

To simulate a peptide on platinum in water, a computer needs rules for how every pair of atoms pulls or pushes. Good rules existed for metals and for molecules, but each was written in its own equations, so nothing said how a metal atom and a water molecule should interact.

The Interface Force Field (IFF) writes the metal in the molecules' equations, then tunes it to measured surfaces.

Built separately



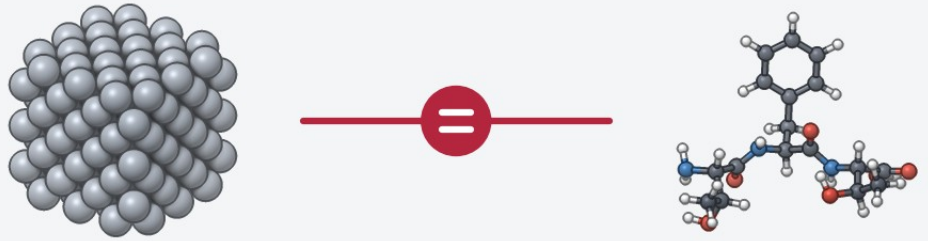
Metal model
its own equations

Molecular model
different equations

No shared rule for metal with molecules

Up to 50% low surface energies in some of the compared metal-only models

Interface Force Field



Metal model
the molecules' equations

Molecular model
unchanged

So metal–molecule pairs follow the same rules

Calibrated to measured lattice size and surface energy, with selected interfaces tested separately

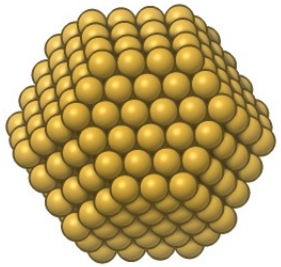
Kanhaiya et al., npj Comput. Mater. (2021) • Heinz et al., Langmuir (2013)

IFF puts solids and molecules in one compatible model

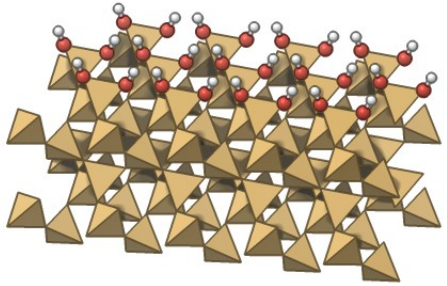
IFF covers solids such as metals, silica, the mineral in bone and teeth, and cement. Each is written in the same style as the standard models for water, proteins and polymers, the long chains in plastics, and its surfaces are built to match real ones.

So one simulation can hold a solid, water and molecules together, and its results can be checked against experiments.

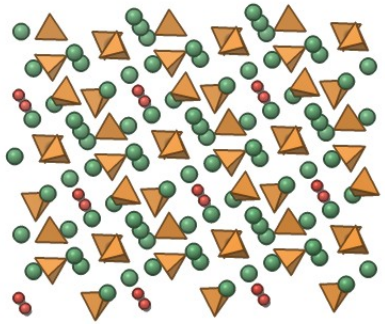
SOLIDS AND THEIR SURFACES



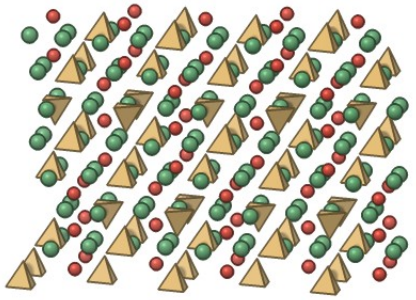
Metals
gold, platinum and other metals



Silica
sand and glass, with OH groups

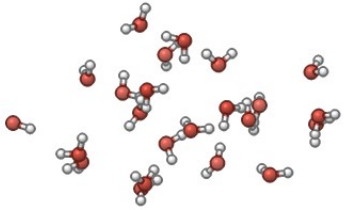


Hydroxyapatite
the mineral in bone and teeth

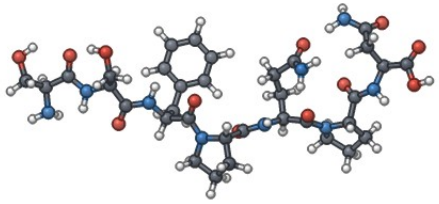


Cement
tricalcium silicate

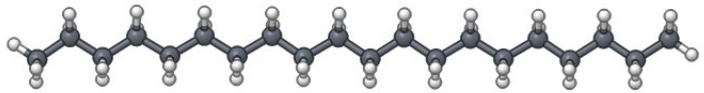
COMBINED WITH STANDARD MOLECULE MODELS SUCH AS AMBER, CHARMM, OPLS-AA AND PCFF



Water



Peptides and proteins



Polymers

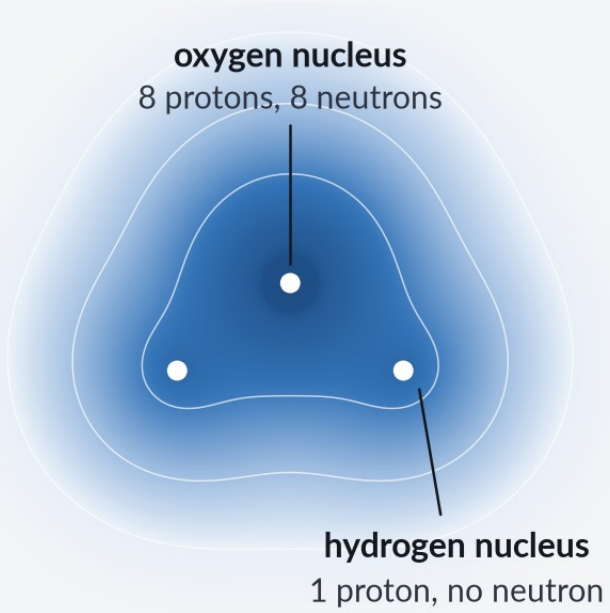
Heinz et al., Langmuir (2013) • Kanhaiya et al., npj Comput. Mater. (2021) • COD 9008463 (gold) • COD 9000775 (quartz) • COD 9011091 (hydroxyapatite) • COD 1540704 (tricalcium silicate) • RDKit

A force field replaces the electron cloud with simple rules

Atoms are held together by their electrons. Quantum chemistry computes the electron cloud directly and is usually much more expensive, so typical calculations hold hundreds or a few thousand atoms. A force field replaces the cloud with balls, springs and small fixed charges, which permits much larger systems, often millions of atoms, and longer runs.

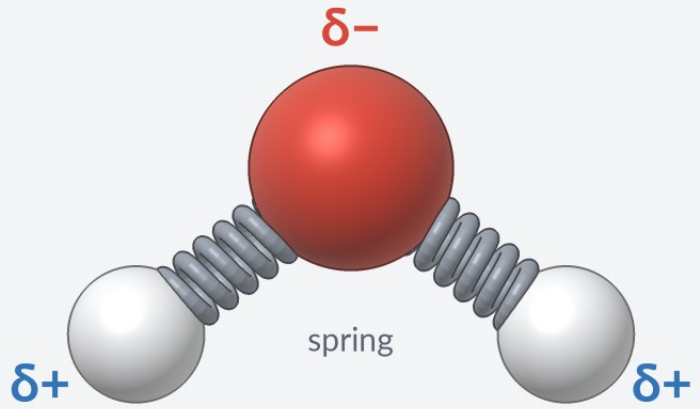
The catch is that the numbers in these rules have to be tuned and checked against experiments.

A cloud of electrons



Quantum chemistry computes this cloud, usually at a high cost

Simple rules instead



Balls, springs and charges often reach millions of atoms

The energy is a sum of five kinds of terms

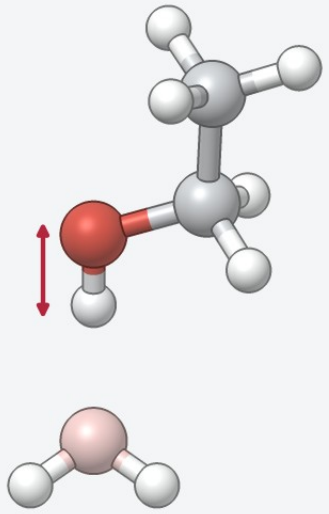
A force field works out the energy of a group of atoms by adding up five simple kinds of terms. Here they act on a real ethanol molecule next to a water molecule. Bonds stretch, angles bend and groups twist around bonds, while atoms that are not bonded feel van der Waals forces and the pull or push of their charges.

The first three keep each molecule in shape. Van der Waals and charges also act between molecules and solids, so they decide what sticks where.

ALONG THE BONDS

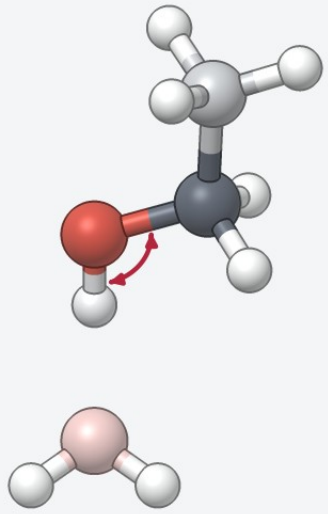
1 Bond stretch

A bond stretches and squeezes like a spring



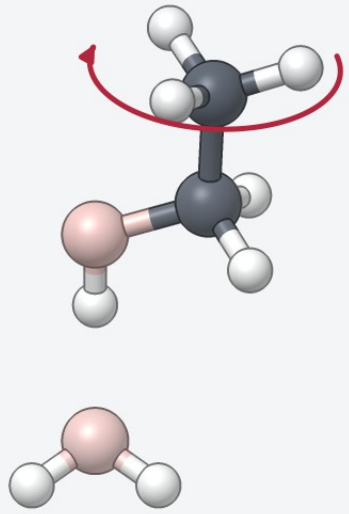
2 Angle bend

The angle between two bonds bends like a spring



3 Twist

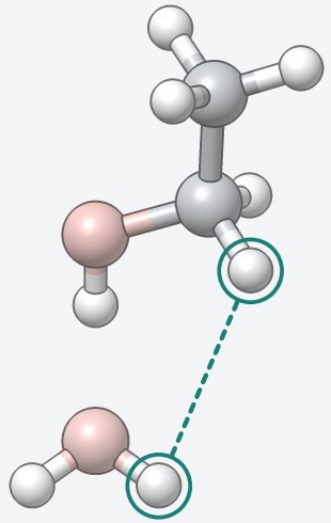
A group of atoms turns around a bond



BETWEEN ATOMS THAT ARE NOT BONDED

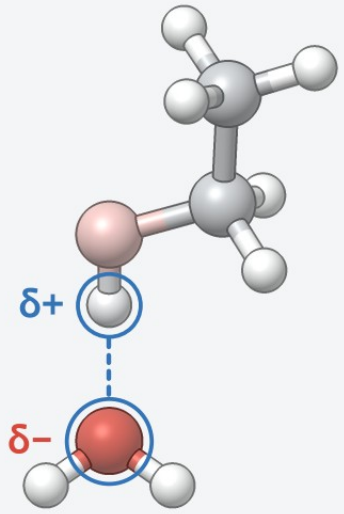
4 Van der Waals

Atoms that are not bonded pull weakly on each other



5 Charges

Opposite partial charges attract each other



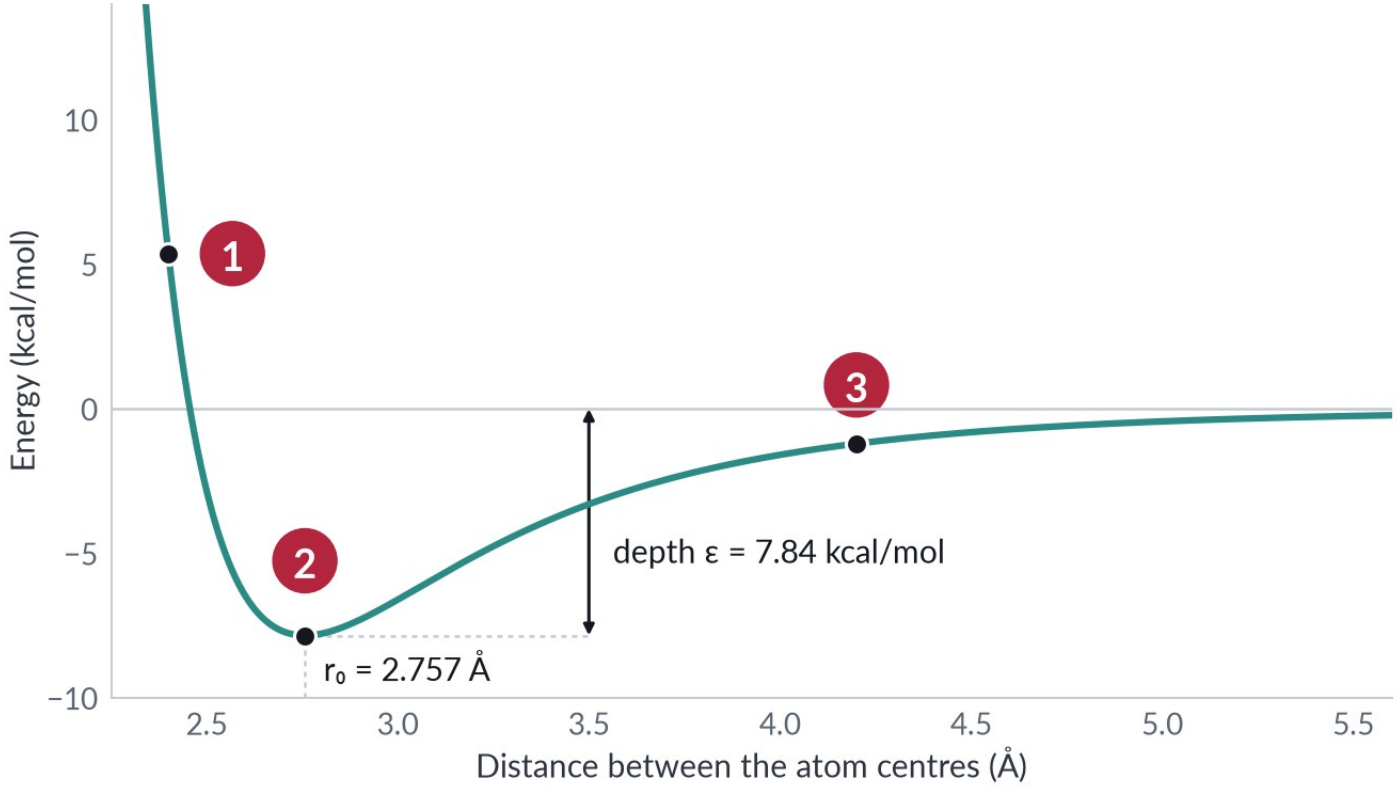
Energy = 1 + 2 + 3 + 4 + 5 added up over every bond, angle, twist and pair of atoms

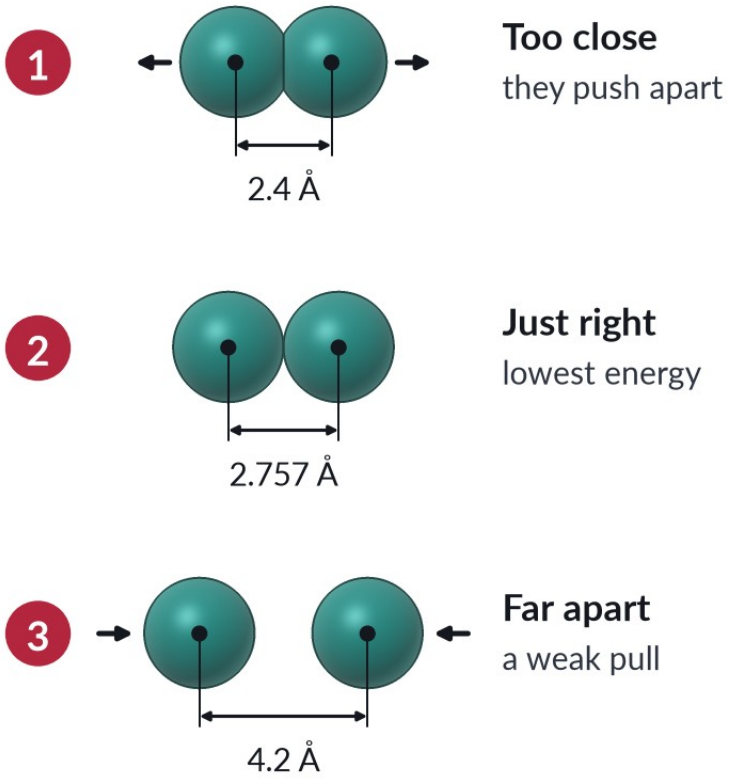
Heinz et al., Langmuir (2013)

Two atoms attract until they get too close

Even neutral atoms pull on each other weakly, and push apart hard when their electron clouds overlap. The Lennard-Jones curve shows this energy against distance for two rhodium atoms, using IFF's published numbers. The lowest point is where the pair would rest.

The distance of the lowest point, r_0, and the depth of the well, ε, set the whole curve.



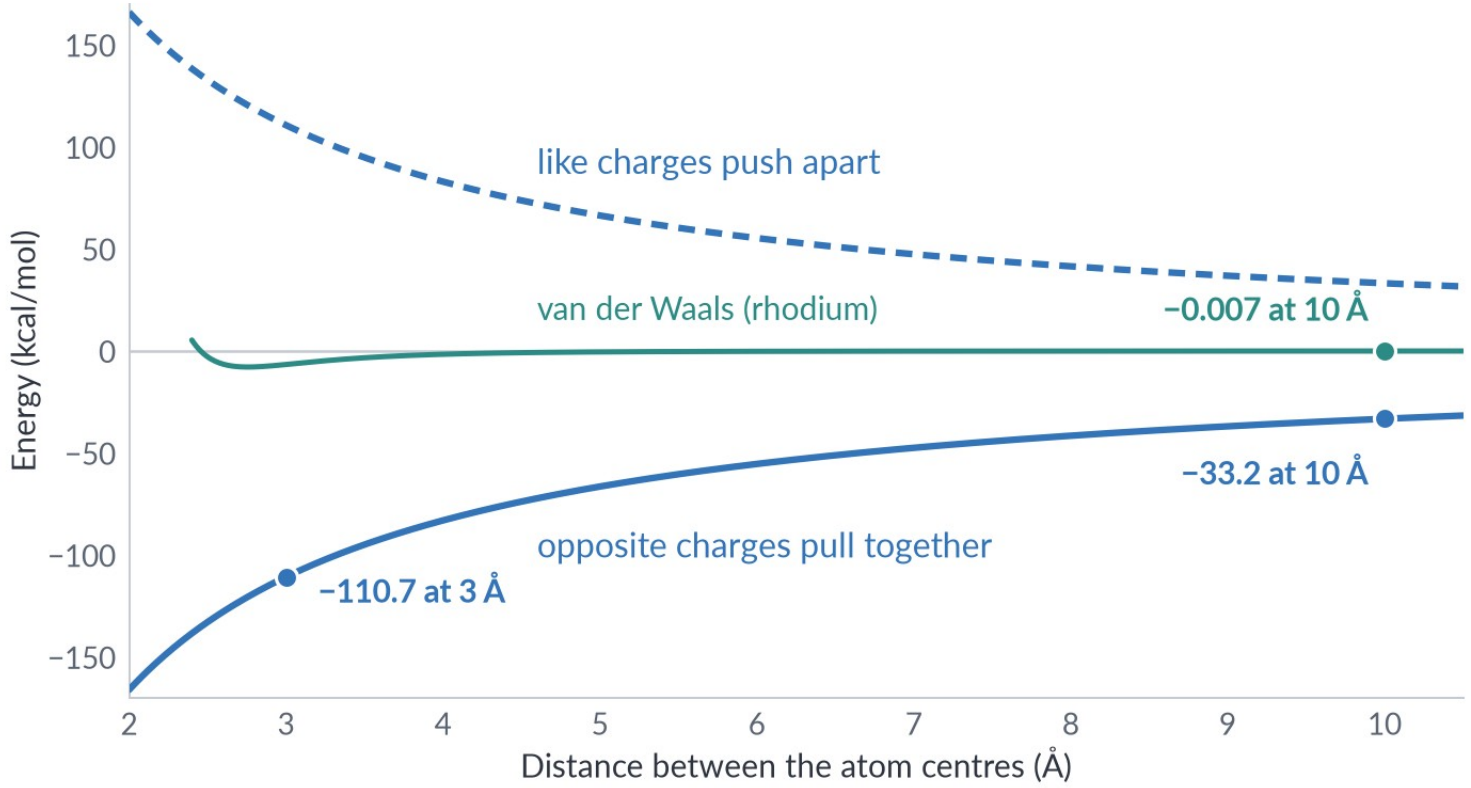


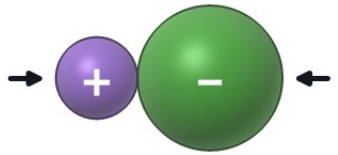
Kanhaiya et al., npj Comput. Mater. (2021)

Charges reach much further than van der Waals

Atoms in a force field also carry charges. Opposite charges pull together and like charges push apart, and the energy fades only slowly with distance. Here the interaction energy of two unit charges in a vacuum, like a sodium and a chloride ion, is drawn on the same axes as the rhodium pair from before.

At 10 Å the energy of two opposite charges is still −33 kcal/mol, while the van der Waals energy has all but vanished.





Opposite charges attract

Na^+ and Cl^-, as in table salt



Like charges repel

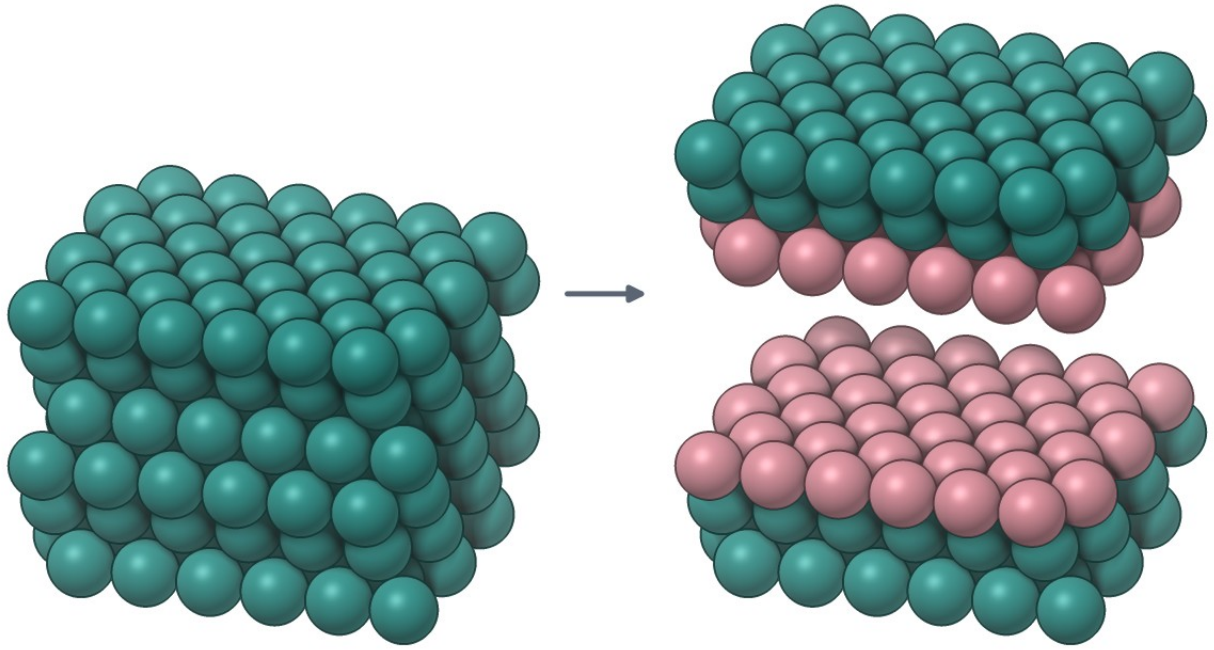
two Na^+ ions

NIST CODATA (2022) • Kanhaiya et al., npj Comput. Mater. (2021) • Calculations for this presentation

IFF tunes rhodium to its measured size and surface energy

Atoms at a surface have lost some neighbours, so making a surface costs energy. Splitting a block makes two new faces, and the energy needed divided by their area is the surface energy. IFF tunes its two rhodium numbers so the simulated block has the measured size and surface energy.

Stiffness was not tuned, so it is a real test, and the stretching stiffness comes out 45% low, 209 against 379 GPa.



One block

Pulled apart

two new faces, each of area A

Surface energy =

energy to pull apart ÷ 2A

Rhodium	
measured	**2.64 ± 0.02 J/m²**
IFF	**2.643 J/m²**

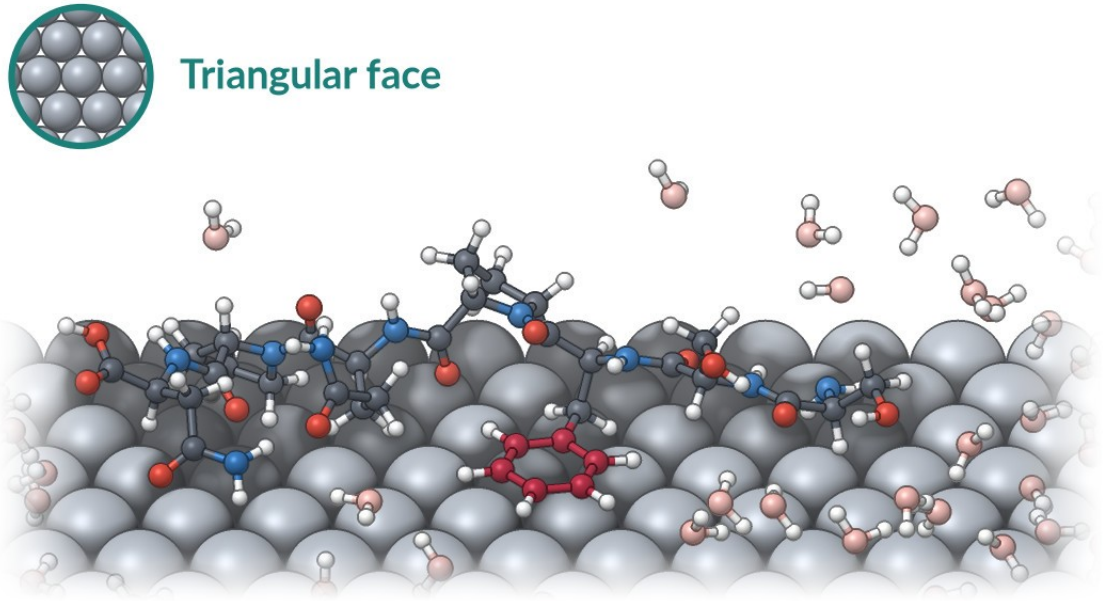
IFF was tuned to match this

Kanhaiya et al., npj Comput. Mater. (2021) • Kanhaiya et al., supplement (2021)

Example 1: The F ring lies flat on platinum's triangular face

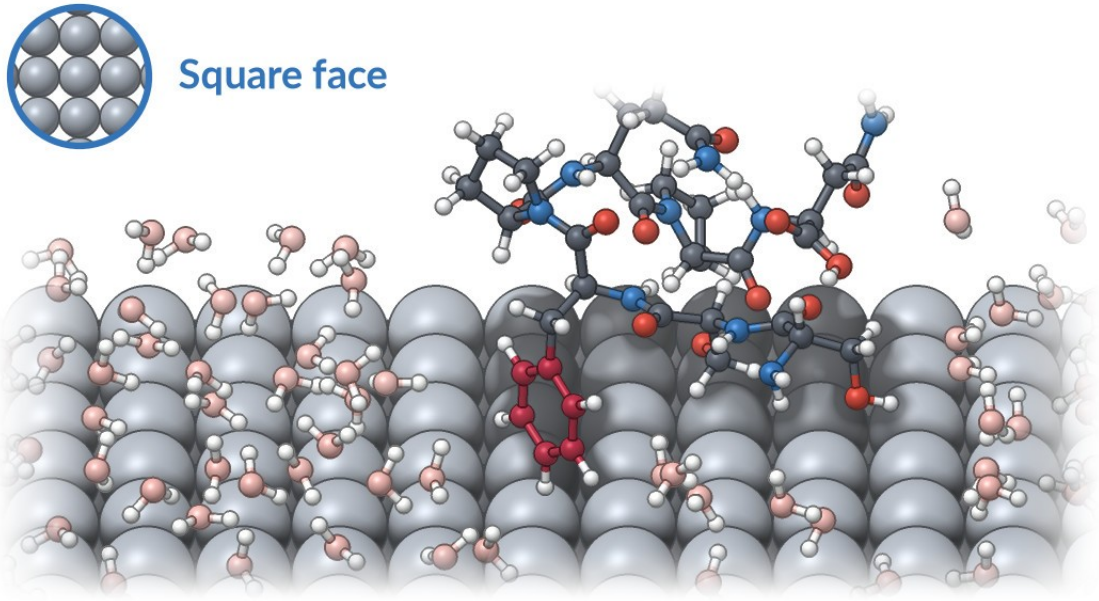
Ruan and colleagues simulated S7, a seven-amino-acid peptide with an F, on both platinum faces in water, using IFF's early metal model. On the triangular face its flat ring lies down and the peptide binds strongly. On the square face the ring stands up, adsorption is less favourable, and water competes more successfully for the surface.

In these syntheses, what mattered was the contrast between the faces, not just how strongly the peptide binds.



Lies flat and binds strongly

its hexagon fits the triangular pattern



Stands up, adsorption is less favourable

computed +3 kcal/mol, so water competes more successfully

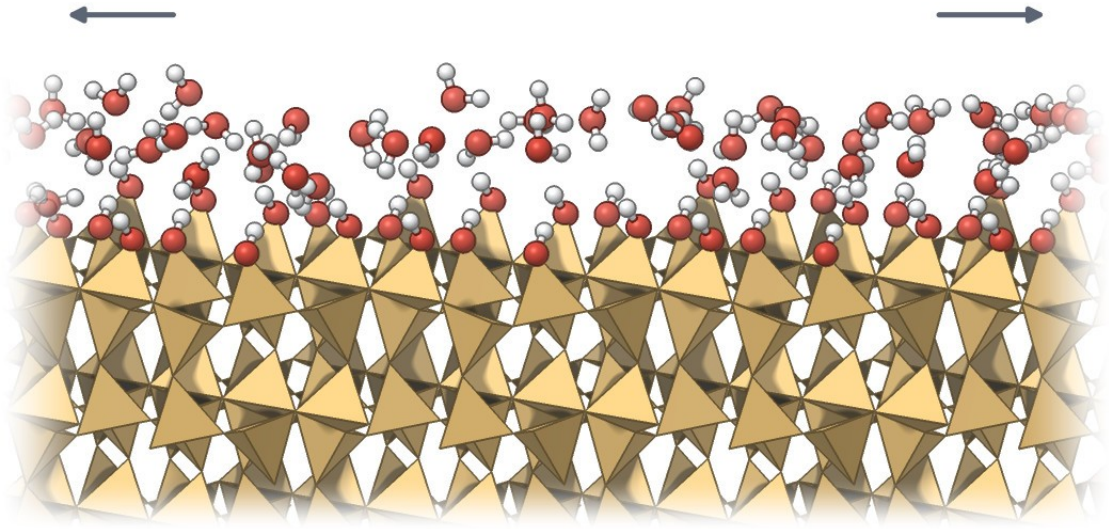
In the lab, every tested peptide version with F grew many tetrahedra, and versions without F grew few

Ruan et al., Nano Lett. (2013) • COD 9008480 (platinum) • RDKit

Example 2: Water spreads or beads up depending on the silica surface

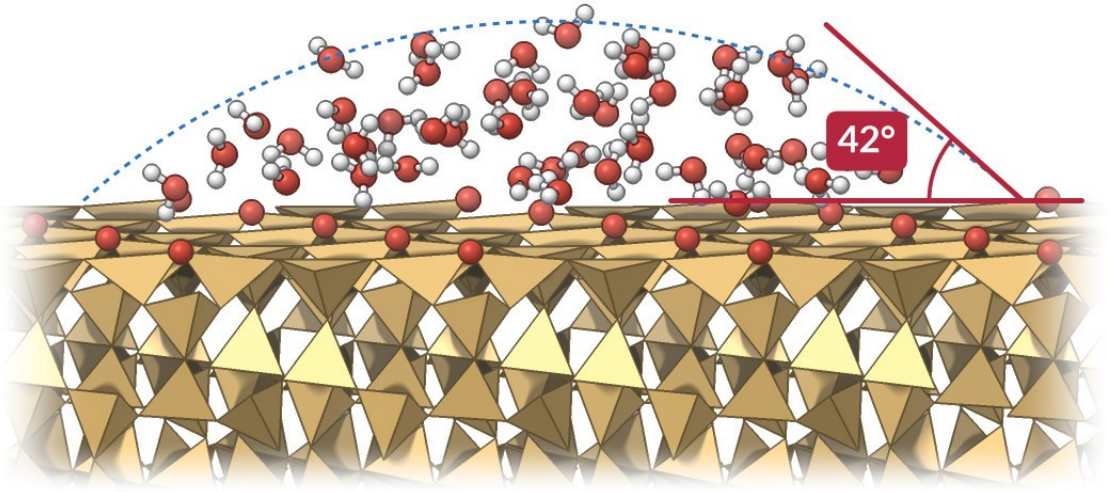
Silica, the main part of sand and glass, carries OH groups on its surface, each an oxygen bonded to a hydrogen. Heating joins neighbouring pairs into Si–O–Si bridges and gives off water. Water spreads flat on the OH-covered surface and beads up on the heated one. Emami and colleagues simulated water on a silica model for each sample, and the angles came within 3° of the measured ones.

The match depends on building each model surface with the same OH density as its real sample.



OH groups on the surface

water spreads flat (0°)



Heating joins OH pairs into Si–O–Si bridges

water beads up (42°)

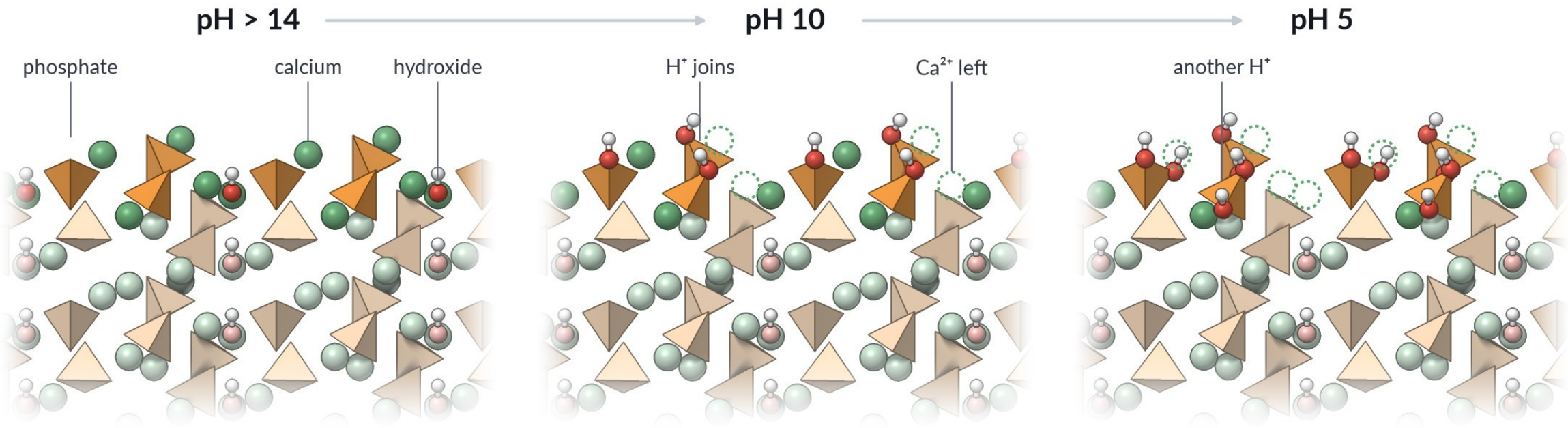
The simulated angles came within 3° of the measured ones on every surface tested, including two in between

Emami et al., Chem. Mater. (2014) • Emami et al., correction (2016) • COD 9009686 (cristobalite)

Example 3: Bone mineral changes its surface with pH

Hydroxyapatite, the mineral in bone and teeth, is made of calcium, phosphate and hydroxide. As the water around it turns more acidic, which means a lower pH, hydrogen ions join the surface phosphates and some calcium leaves. Lin and Heinz built a model surface for each pH.

The surface a molecule meets depends on pH, so a model must be built for the pH of the experiment.



The bare crystal

PO_4^{3-} on top, stable only at very high pH

One H on each phosphate

HPO_4^{2-}, with hydroxide and some Ca^{2+} gone

Two H on each phosphate

$H_2PO_4^-$, with more Ca^{2+} gone

The simulated mineral gives off 619 mJ/m² as water wets it at pH 7 (533 if all faces count equally), and experiments find 600–700

Lin and Heinz, J. Phys. Chem. C (2016) • COD 9011091 (hydroxyapatite)

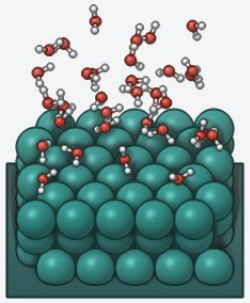
The IFF Agent carries one material through eight steps

The IFF Agent is the author's described workflow for building and checking an IFF model, one material at a time. Its first four steps decide what the result must show before anything is computed, including which measurements may tune the model and which are kept back as tests, so a test is never a number the model was tuned to.

The IFF Agent is a workflow around IFF, not part of the force field, and this presentation claims no results from it.

Set up decide what the result must show

1 Ask



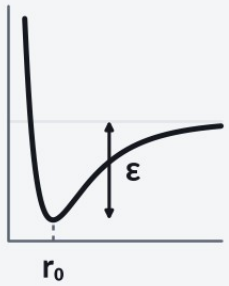
rhodium faces in water

2 Gather



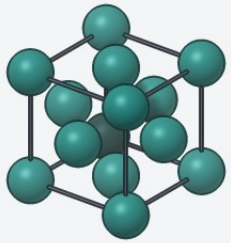
measured size, surface energy and stiffness

3 Choose



a 12-6 pair curve with r_0 and ε

4 Plan



tune size and surface energy, test stiffness

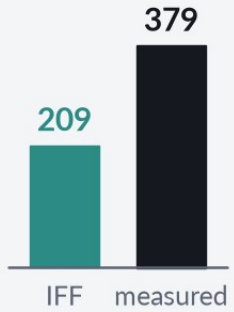
Run and check compute, test, then decide

5 Fit



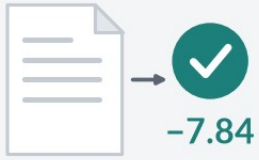
r_0 = 2.757 Å
ε = 7.84 kcal/mol

6 Test



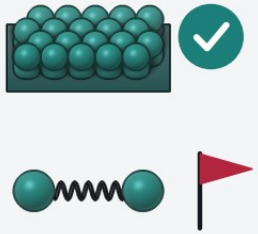
stretching is 209 vs 379 GPa, 45% low

7 Read back



energy at r_0 is −7.84, not 0

8 Decide



use for surfaces, flag stiffness

Every number comes from rhodium's published values, and nothing here is a new fit or result

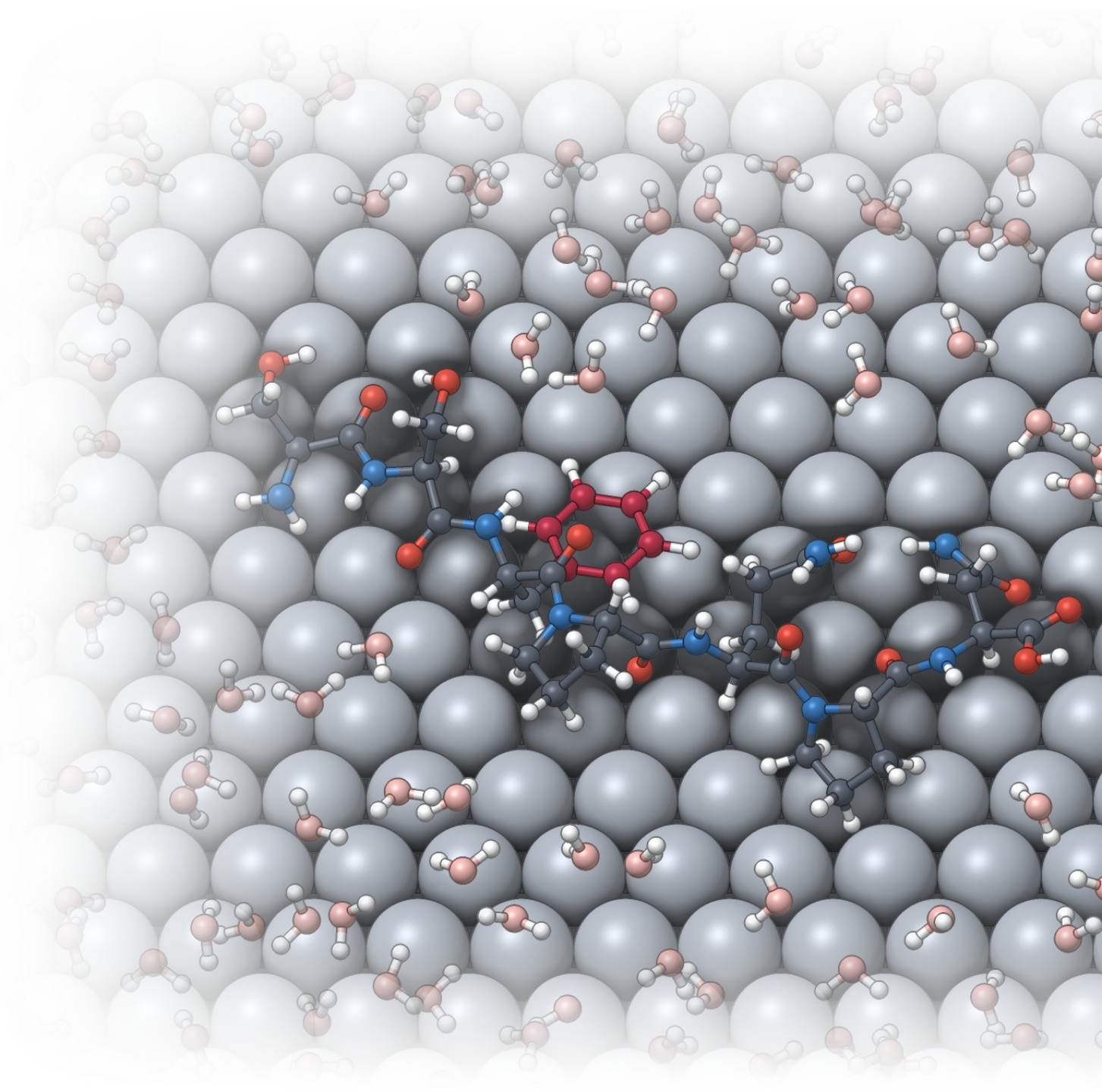
THE ANSWER · CRYSTAL SHAPE

A flat ring on a triangular face explains the crystal shape

Swapping L for F shifted platinum crystals from cubes toward tetrahedra. In the simulations the F ring lies flat on triangular faces but not on square ones. In these syntheses, binding slows the growth of the triangular faces, so they end up covering the crystal.

> IFF puts metals, minerals, water and molecules into one model that is checked against experiments. With it, researchers can test on a computer which molecules stick to which surfaces before anything is made, from catalysts and medicine to bone and cement.

Thank you. Questions?



Ruan et al., Nano Lett. (2013) • Heinz et al., Langmuir (2013) • Kanhaiya et al., npj Comput. Mater. (2021)